THE MINIMAL PRIMES OF THE STRICT HENSELIZATION OF A TENSOR PRODUCT OF STRICTLY HENSELIAN LOCAL ALGEBRAS CORRESPOND TO PAIRS OF MINIMAL PRIMES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The seven references inside the counted proof span are recorded as non-load-bearing navigation pointers; they name structural facts about strict henselization and minimal primes whose applications stay inside the counted proof span.

Allowed standard prerequisites (editorial).

- Strict henselization of a local ring as a filtered colimit of etale neighbourhoods and its residue field. used throughout the counted proof span. Is the construction whose minimal primes are being described.
- Minimal primes of a tensor product of algebras over an algebraically closed field. used in the counted proof span. Supplies the pair-of-minimal-primes description that is transported to C.

Source reference. Original lemma 00HS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 654–654. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 06DF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8179–8179. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 00GU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 620–620. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 05WS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7596–7596. Cross-chapter navigation pointer into

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the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 00EA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 527–527. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 31453: sources/stacks-more-algebra/pinned-original-source.tex, source lines 31453–31453. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let k be an algebraically closed field. Let A, B be strictly henselian local k-algebras with residue field equal to k. Let C be the strict henselization of $A \otimes_k B$ at the maximal ideal $\mathfrak{m}_A \otimes_k B + A \otimes_k \mathfrak{m}_B$. Then the minimal primes of C correspond 1-to-1 to pairs of minimal primes of A and B.*

Proof. First note that a minimal prime $\mathfrak{r}$ of C maps to a minimal prime $\mathfrak{p}$ in A and to a minimal prime $\mathfrak{q}$ of B because the ring maps $A \to C$ and $B \to C$ are flat (by going down for flat ring map Algebra, Lemma 00HS). Hence it suffices to show that the strict henselization of $(A/\mathfrak{p} \otimes_k B/\mathfrak{q})_{\mathfrak{m}_A \otimes_k B + A \otimes_k \mathfrak{m}_B}$ has a unique minimal prime ideal. By Algebra, Lemma 05WS the rings $A/\mathfrak{p}$, $B/\mathfrak{q}$ are strictly henselian. Hence we may assume that A and B are strictly henselian local domains and our goal is to show that C has a unique minimal prime. By Lemma 31453 the integral closure A' of A in its fraction field is a normal local domain with residue field k. Similarly for the integral closure B' of B into its fraction field. By Algebra, Lemma 06DF we see that $A' \otimes_k B'$ is a normal ring. Hence its localization

$$R = (A' \otimes_k B')_{\mathfrak{m}_{A'} \otimes_k B' + A' \otimes_k \mathfrak{m}_{B'}}$$

is a normal local domain. Note that $A \otimes_k B \to A' \otimes_k B'$ is integral (hence gong up holds – Algebra, Lemma 00GU) and that $\mathfrak{m}_{A'} \otimes_k B' + A' \otimes_k \mathfrak{m}_{B'}$ is the unique maximal ideal of $A' \otimes_k B'$ lying over $\mathfrak{m}_A \otimes_k B + A \otimes_k \mathfrak{m}_B$. Hence we see that

$$R = (A' \otimes_k B')_{\mathfrak{m}_A \otimes_k B + A \otimes_k \mathfrak{m}_B}$$

by Algebra, Lemma 00EA. It follows that

$$(A \otimes_k B)_{\mathfrak{m}_A \otimes_k B + A \otimes_k \mathfrak{m}_B} \longrightarrow R$$

is integral. We conclude that R is the integral closure of $(A \otimes_k B)_{\mathfrak{m}_A \otimes_k B + A \otimes_k \mathfrak{m}_B}$ in its fraction field, and by Lemma 31453 once again we conclude that C has a unique prime ideal. □